



Time: 5 minutes

Outcome: Connect Elasticity and Total Revenue.

Difficulty: Intermediate

Elasticity and Total Revenue

THE CORE IDEA

Total revenue equals price multiplied by quantity, and the effect of a price change depends on demand elasticity. A point on demand supplies the price and quantity needed to calculate revenue. At unit elasticity the price and quantity percentage effects offset, marking the revenue peak on a standard linear demand curve.

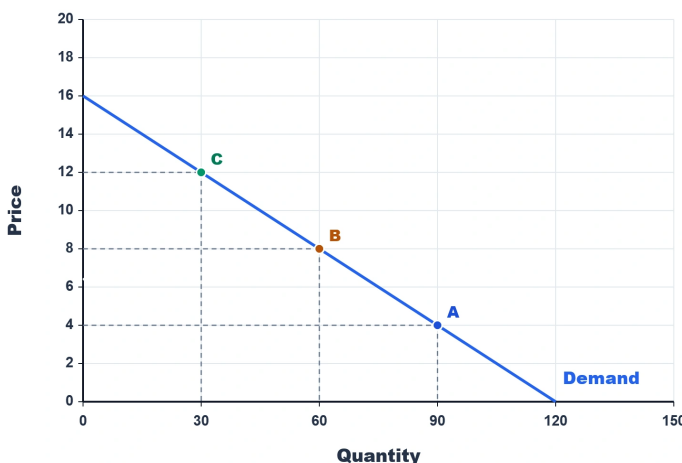
HOW TO RECOGNIZE IT

- Calculate total revenue as price multiplied by quantity at each relevant point.
- If price and total revenue move in opposite directions, demand is elastic over that range.
- If price and total revenue move in the same direction, demand is inelastic over that range.
- At unit elasticity, a small price change leaves total revenue approximately unchanged.

WATCH OUT

A common mistake is to compute elasticity without checking the associated total-revenue change. With elastic demand, price and total revenue move in opposite directions; with inelastic demand, they move together.

WORKED EXAMPLE: ELASTICITY AND REVENUE



At point C, price is \$12 and quantity is 30, so total revenue is $\$12 \times 30 = \360 . At point B, price is \$8 and quantity is 60, so revenue is \$480. Moving from C to B lowers price but raises total revenue by \$120, indicating elastic demand over that segment.

CHECK YOURSELF

Cruise-cabin price falls from \$3,200 to \$2,880 while quantity rises from 1,500 to 1,900. Calculate midpoint elasticity and both revenue totals. Is demand elastic over this range, and does revenue rise or fall?

READY?

Return to the game and master this concept.